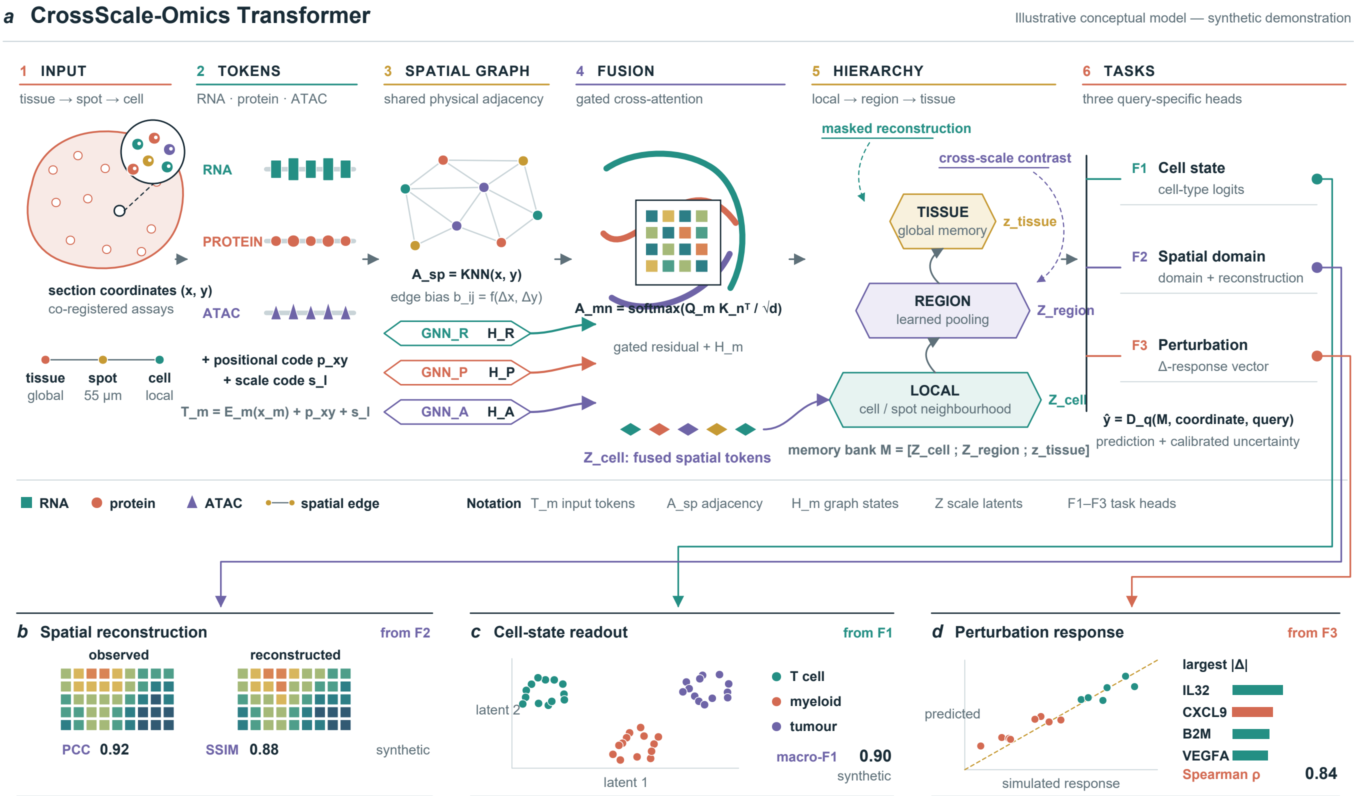
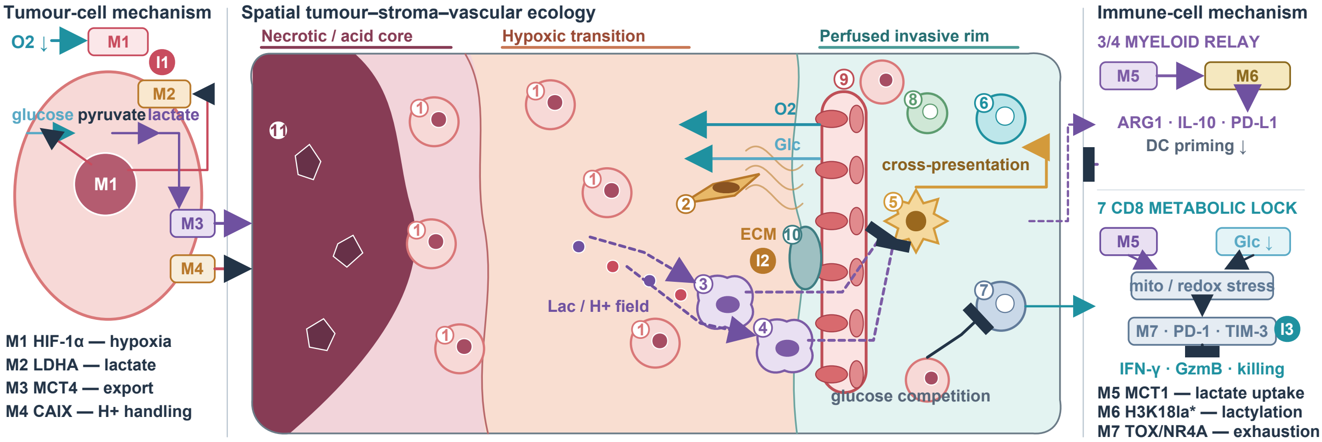


Illustrative conceptual model; synthetic data and metrics; no empirical performance claims.

SPATIAL COUPLING OF HYPOXIA, LACTATE AND IMMUNE EXHAUSTION

Tumour source → acidic myeloid niche → CD8 T-cell metabolic lock

ILLUSTRATIVE HYPOTHETICAL MECHANISM · NOT DATA



CELL / STRUCTURE KEY 1 tumour 2 CAF / ECM 3 TAM 4 MDSC 5 DC 6 CD8 effector 7 CD8 exhausted 8 Treg 9 vessel / endothelium 10 pericyte 11 necrotic debris Lac H+ O2 Glc

THREE TESTABLE INTERVENTION WINDOWS

I1 Reduce tumour source

LDHA / MCT4 blockade

Predicted: lactate export ↓

I2 Recondition spatial niche

buffer H+ · loosen CAF/ECM gate myeloid MCT1

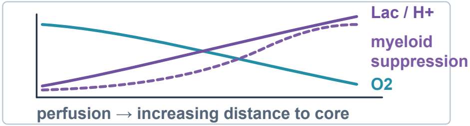
Predicted: suppression ↓

I3 Restore effector fitness

PD-1 release + mitochondrial support

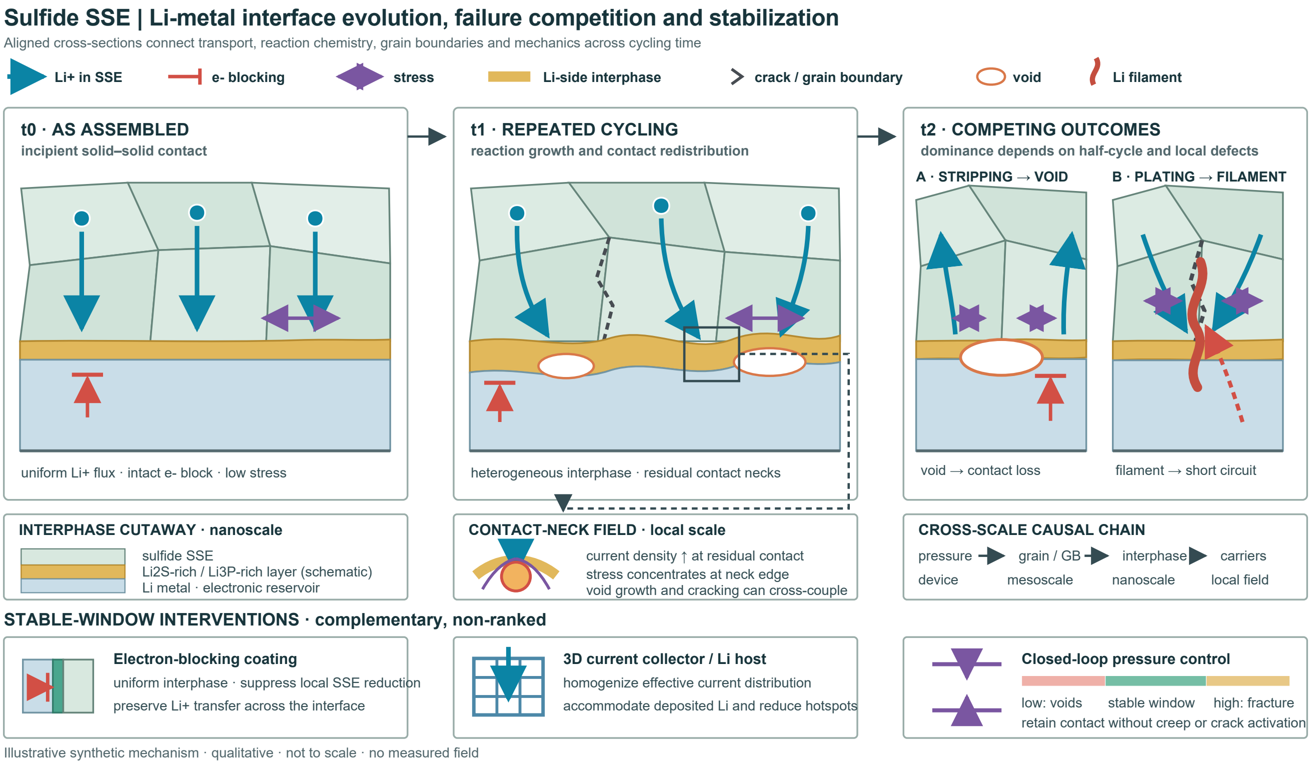
Predicted: cytotoxicity ↑

SCHEMATIC SPATIAL GRADIENT · NOT DATA



Arrow = promotion · bar = inhibition · dashed = proposed / context-dependent

Illustrative hypothetical mechanism · microplot not data


Sulfide SSE | Li-metal interface evolution, failure competition and stabilization
Aligned cross-sections connect transport, reaction chemistry, grain boundaries and mechanics across cycling time
Li+ in SSE
e- blocking
stress
Li-side interphase
crack / grain boundary
void
Li filament
t0 · AS ASSEMBLED
incipient solid–solid contact
uniform Li+ flux · intact e- block · low stress
t1 · REPEATED CYCLING
reaction growth and contact redistribution
heterogeneous interphase · residual contact necks
t2 · COMPETING OUTCOMES
dominance depends on half-cycle and local defects
A · STRIPPING → VOID
B · PLATING → FILAMENT
void → contact loss
filament → short circuit
INTERPHASE CUTAWAY · nanoscale
sulfide SSE
Li2S-rich / Li3P-rich layer (schematic)
Li metal · electronic reservoir
CONTACT-NECK FIELD · local scale
current density ↑ at residual contact
stress concentrates at neck edge
void growth and cracking can cross-couple
CROSS-SCALE CAUSAL CHAIN
pressure
device
grain / GB
mesoscale
interphase
nanoscale
carriers
local field
STABLE-WINDOW INTERVENTIONS · complementary, non-ranked
Electron-blocking coating
uniform interphase · suppress local SSE reduction
preserve Li+ transfer across the interface
3D current collector / Li host
homogenize effective current distribution
accommodate deposited Li and reduce hotspots
Closed-loop pressure control
low: voids
stable window
high: fracture
retain contact without creep or crack activation
Illustrative synthetic mechanism · qualitative · not to scale · no measured field

a **Illustrative warming-pressure states**

S1
Guarded warming +1.4 °C
lower coupling load

S2
Persistent warming +2.1 °C
intermediate coupling load

S3
High overshoot +3.2 °C
higher coupling load + lag

Illustrative coordinates only — not assessed thresholds or forecasts

b **Coupled spatial network**

numbered anchors carry geography · outer arcs carry system-wide feedback · E1–E12 carry directed coupling

EDGE KEY · directed influence on cascade tendency

E1 GrIS → AMOC + Freshwater weakening	E5 AMOC → AMZ ± Rainfall redistribution	E9 PF → AMZ + Carbon–warming bridge
E2 AMOC → GrIS − Regional cooling	E6 AMZ → AMOC ± Moisture recycling	E10 PF → CR + Ocean-warming bridge
E3 AIS → AMOC + Southern freshening	E7 AMOC → PF ± Poleward heat	E11 AMZ → AMZ + Fire–drying feedback
E4 AMOC → AIS ± Heat redistribution	E8 AMOC → CR + Marine heat exposure	E12 CR → CR + Recovery-capacity loss

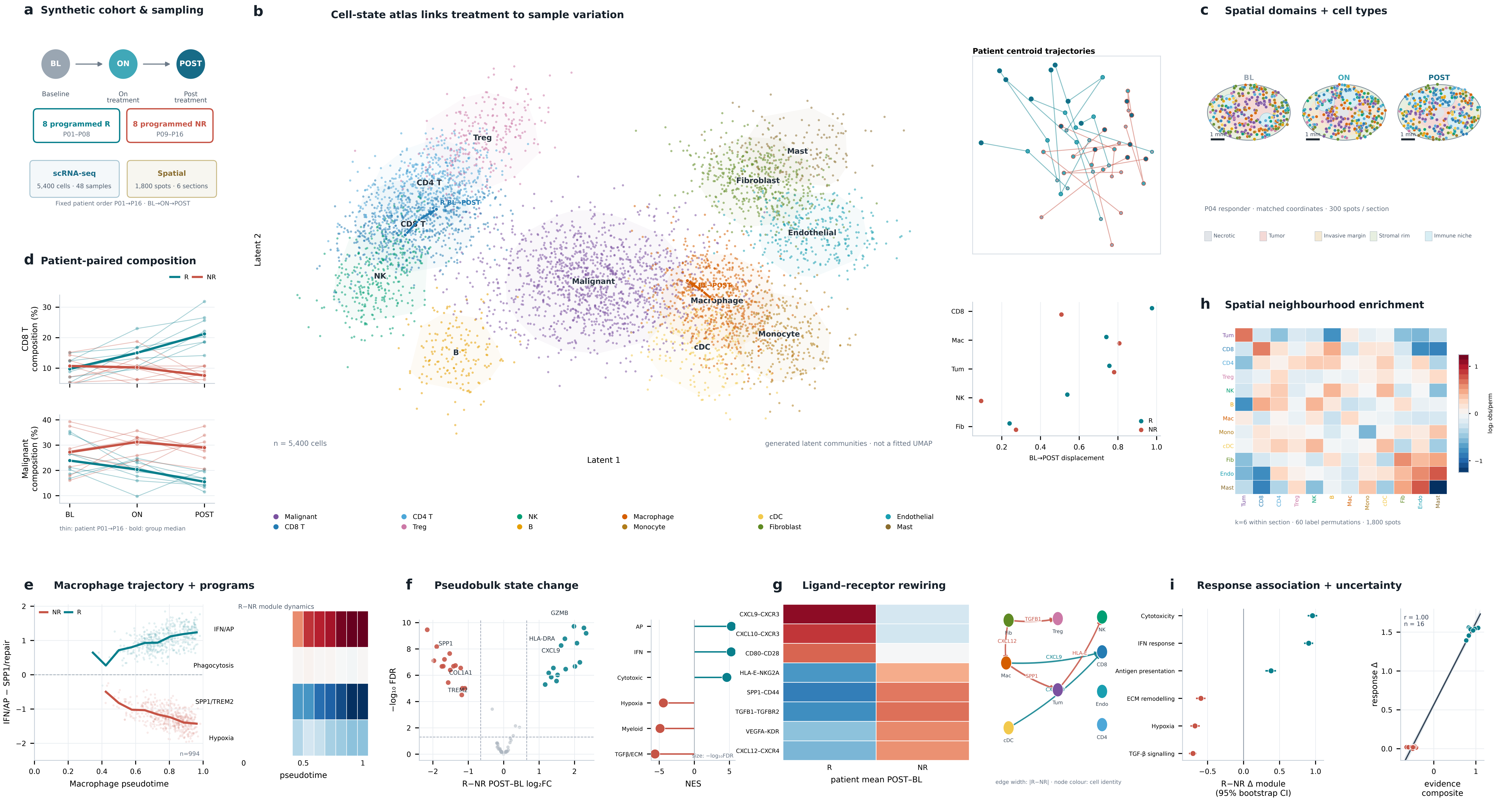
+ amplifying/destabilizing − damping/temporarily stabilizing ± context-dependent

Conceptual synthesis; signs and strengths vary by state, model and forcing pathway.

A synthetic treatment-aware single-cell and spatial tumor ecosystem atlas

SYNTHETIC DEMONSTRATION DATA

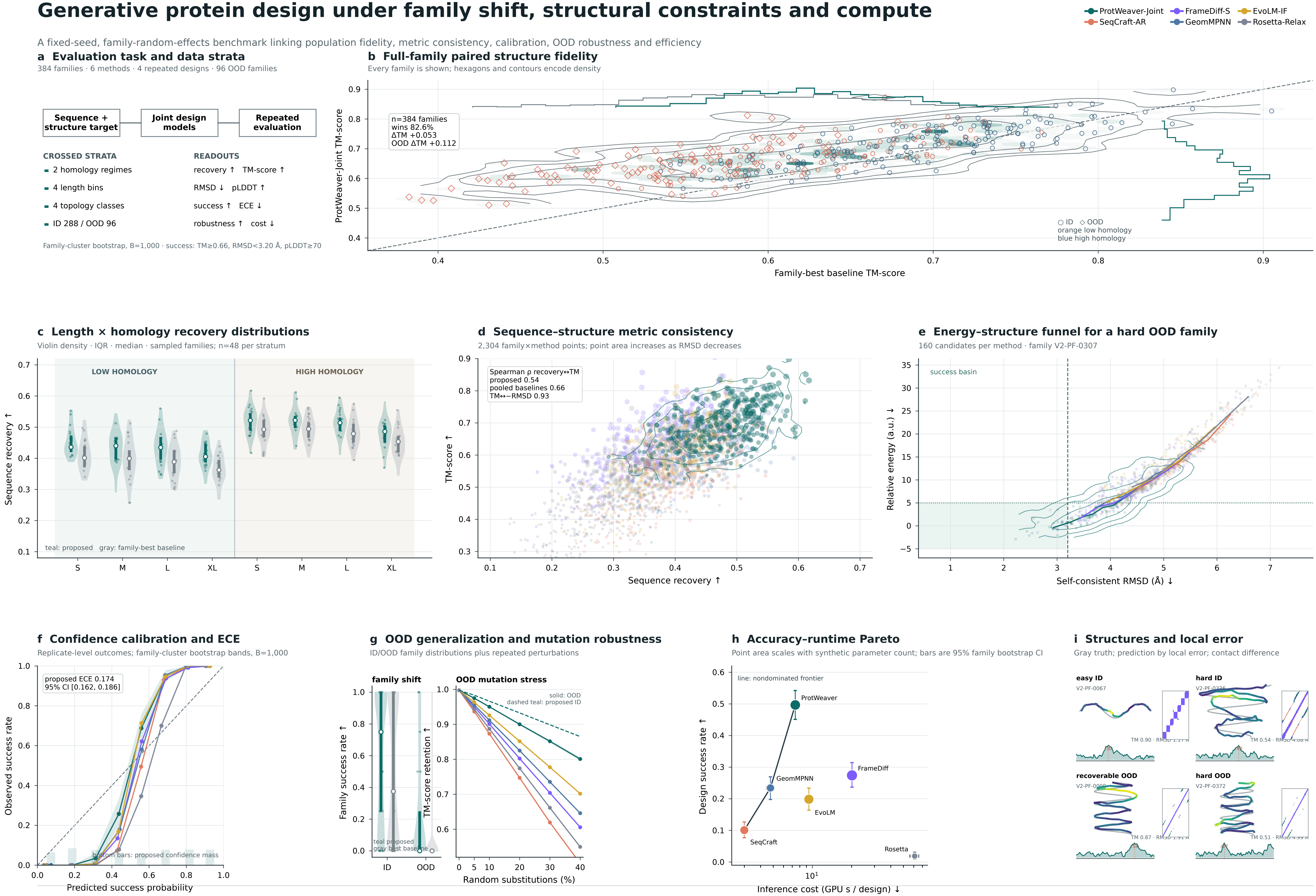
Longitudinal cell states → tissue domains → paired remodelling → molecular programs → communication → neighbourhoods → response uncertainty



Synthetic demonstration data · 5,400 cells + 1,800 spatial spots · 16 virtual patients · fixed seed 30527 · no patient-derived observations

Generative protein design under family shift, structural constraints and compute

A fixed-seed, family-random-effects benchmark linking population fidelity, metric consistency, calibration, OOD robustness and efficiency

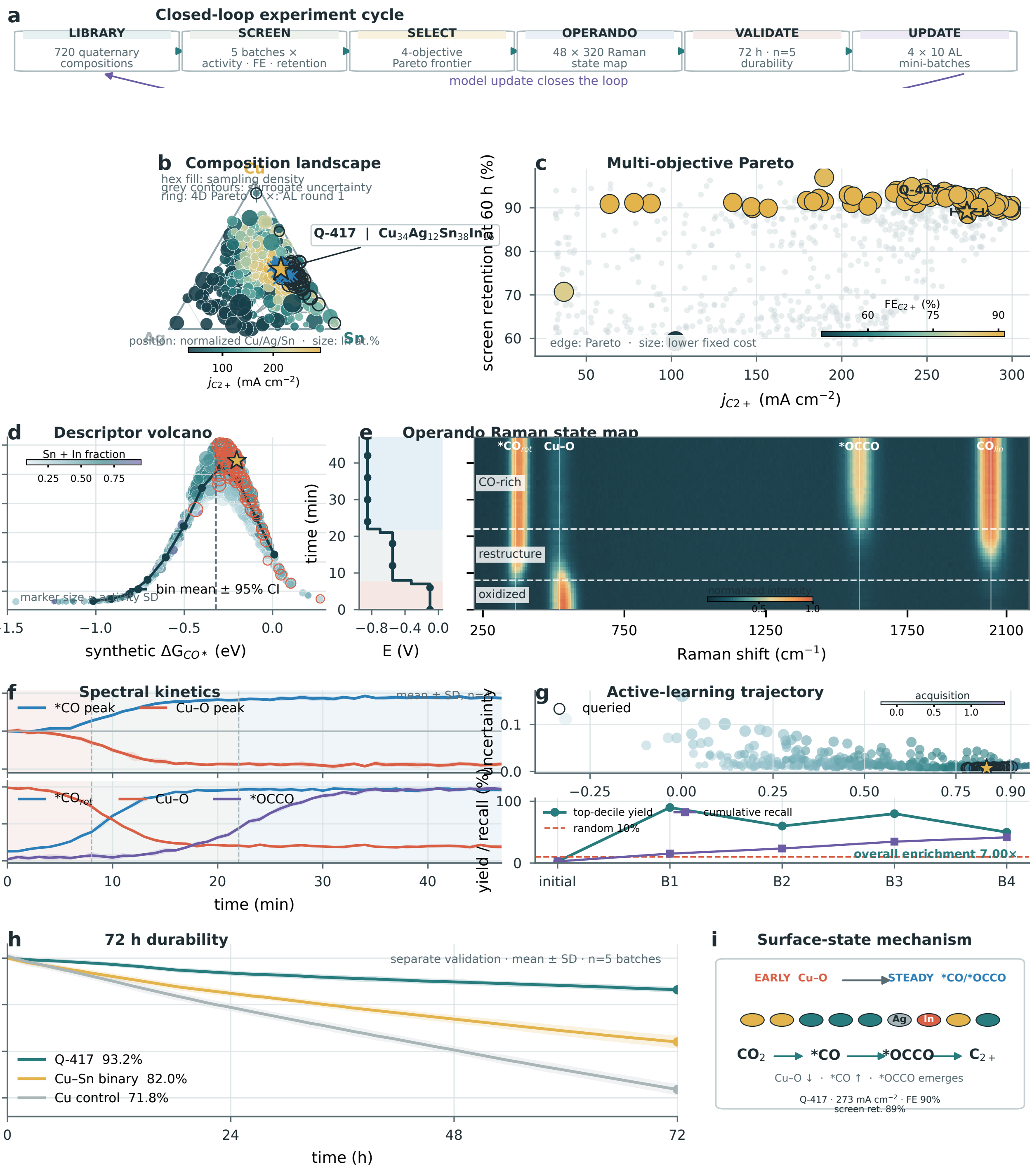


Synthetic benchmark for visualization demonstration

seed 30620260828 · n=384 families × 6 methods × 4 repeats · 95% CI: family-cluster bootstrap B=1,000

Closed-loop discovery links composition, operando state and 72 h durability

Cu–Ag–Sn–In CO_2RR | fixed seed 307 | Q-417 = $Cu_{34}Ag_{12}Sn_{38}In_{16}$

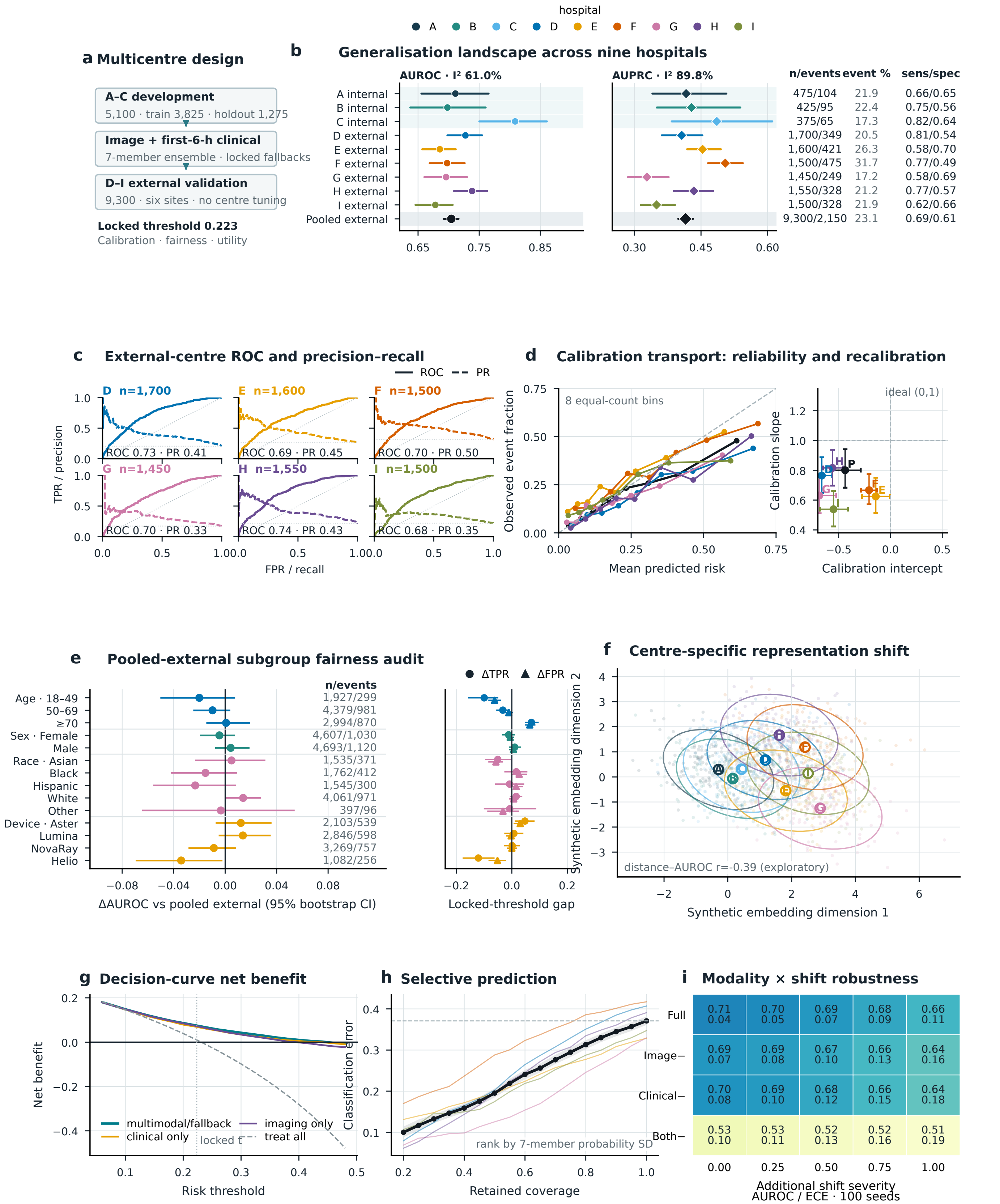


screen n=720, repeats n=5 | operando n=4 | bootstrap=1,000 | AL round 1: 80 + 4×10

Synthetic catalyst-screening demonstration data

Multicentre external validation of a synthetic multimodal ICU risk model

Synthetic multi-centre validation data for visualization demonstration



Synthetic throughout. Centre/subgroup CIs: 400 bootstraps; pooled analyses preserve centre×outcome strata; calibration CIs: 240.

Threshold fixed internally and unchanged in D–I. Predictive uncertainty = 7-member probability SD; NOT CLINICAL EVIDENCE.